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Candidate Number

HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION MOCK

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.
- (2) This paper consists of TWO sections, A and B.
- (3) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (4) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (5) Unless otherwise specified, all working must be clearly shown.
- (6) Unless otherwise specified, numerical answers must be exact.
- (7) The diagrams in this paper are not necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$
$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$	$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$	
$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$	

SECTION A (50 marks)

1. Let $f(x) = \sin^2 x + x \cos x$ for all real numbers x .

(a) Express $f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)$ in terms of h . (2 marks)

(b) Define $g(x) = e^{f(x)}$ for all real numbers x . Using first principles, find $g'\left(\frac{\pi}{2}\right)$. (4 marks)

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2. (a) Prove that $\frac{\cos x + \cos 2x - \cos 3x - \cos 4x}{\sin x + \sin 2x + \sin 3x + \sin 4x} = \tan x$, where $\sin x + \sin 2x + \sin 3x + \sin 4x \neq 0$. (2 marks)
- (b) Prove that $\frac{\sin x + \sin 2x + \sin 3x + \sin 4x}{\cos x + \cos 2x + \cos 3x + \cos 4x} = \tan \frac{5x}{2}$, where $\cos x + \cos 2x + \cos 3x + \cos 4x \neq 0$. (1 mark)
- (c) Solve $\frac{\cos \theta + \cos 2\theta - \cos 3\theta - \cos 4\theta}{\sin \theta + \sin 2\theta + \sin 3\theta + \sin 4\theta} = \frac{\sin \theta + \sin 2\theta + \sin 3\theta + \sin 4\theta}{\cos \theta + \cos 2\theta + \cos 3\theta + \cos 4\theta}$ for $-\pi \leq \theta \leq \pi$. (3 marks)

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3. (a) Using integration by parts, prove that

$$\int e^{-x}[f(x) - f'(x)] dx = -e^{-x}f(x) + C,$$

where C is a constant of integration.

(2 marks)

(b) Prove that $\frac{x^n + nx^{n-1} + x + 1}{(x + x^n)^2} = \frac{1}{x + x^n} - \frac{d}{dx} \left(\frac{1}{x + x^n} \right)$ for any positive integer $n \geq 2$.

(2 marks)

(c) Hence, evaluate $\int_5^8 e^{-x} \cdot \frac{x^3 + 3x^2 + x + 1}{(x + x^3)^2} dx$.

(2 marks)

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4. Given that $\vec{v}_1 = (4, 3)$ and $\vec{v}_1 \cdot \vec{v}_2 = 0$. Let $\vec{v}_1 \cdot \vec{v}_3 = \cos \theta$, where θ is the angle between $\vec{v}_1$ and $\vec{v}_3$. It is given that $\vec{v}_2 \cdot \vec{v}_3 > 0$.

(a) Express $\vec{v}_2 \cdot \vec{v}_3$ in terms of $\|\vec{v}_2\|$ and θ . (3 marks)

(b) If $\|\vec{v}_2\| = \frac{1}{5}$ and $\theta = \frac{\pi}{6}$, find the distance between $\vec{v}_2$ and $\vec{v}_3$. (3 marks)

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(b) Hence, find the value of $\sum_{k=10}^{90} (5 + 4k) \cdot 3^{2k+1}$. (2 marks)

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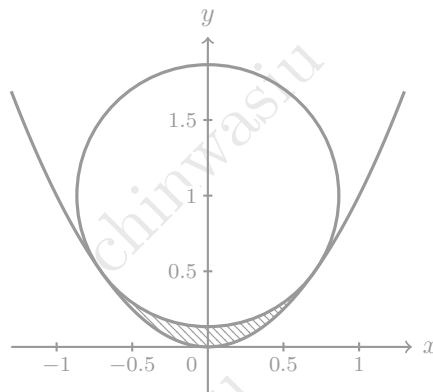
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7. An inner bowl B_1 and an outer bowl B_2 are placed concentrically to form a double-walled container with a cavity in between.

The inner surface of B_1 and the outer surface of B_2 are generated by revolving the curves C_1 and C_2 about the y -axis respectively, where:

$$C_1 : y = x^2$$

$$C_2 : x^2 + (y - 1)^2 = \frac{3}{4} \quad \text{for } y \leq 1$$



It is given that C_1 and C_2 touch each other tangentially at $y = \frac{1}{2}$, sealing the top edge of the cavity.

- Find the total capacity of the cavity between the two bowls. (3 marks)
- Water is poured into the empty cavity between the bowls at a constant rate of $\frac{\pi}{1280}$ cubic units per second. Let h be the depth of water in the cavity at time t seconds (where $0 < h \leq \frac{1}{2}$).
 - Express the volume of water $V(h)$ in the cavity in terms of h . (2 marks)
 - Find the rate of change of the water depth $\frac{dh}{dt}$ when the water depth is $h = \frac{1}{4}$. (2 marks)

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SECTION B (50 marks)

9. Let $f(x) = \frac{x^3+x^2-5x+3}{x^2+3}$ for all real numbers x . Let C be the curve $y = f(x)$.

- (a) Find the asymptote(s) of C . (2 marks)
- (b) Find the point(s) of inflexion of C . (3 marks)
- (c) Sketch C . (3 marks)
- (d) (i) Find the equation of the tangent L to C at $x = -\sqrt{3}$. (2 marks)
- (ii) Prove that $L \geq f(x)$ for all real numbers x . (3 marks)

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10. (a) (i) Using integration by parts, find $\int \ln x \, dx$. (2 marks)

(ii) Let $f(x)$ be a continuous function satisfying $f(-x) = f(x)$ for all real numbers x .
Prove that

$$\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx \quad (2 \text{ marks})$$

(b) Let $t > 1$. Define $I(t) = \int_{1/t}^t \frac{\ln\left(\frac{1+x^2}{x^2}\right)}{x(1+x^2)} \, dx$.

(i) Prove that $I(t) = \int_{1/t}^t \frac{x \ln(1+x^2)}{1+x^2} \, dx$. (2 marks)

(ii) Show that

$$I(t) = \int_0^{\ln t} \left(\ln(1+e^{2x}) - \frac{2x}{1+e^{2x}} \right) \, dx \quad (3 \text{ marks})$$

(iii) Using the results of (a)(i) and (b)(ii), evaluate $\int_{1/e}^e \frac{\ln\left(\frac{1+x^2}{x^2}\right)}{x(1+x^2)} \, dx$ (2 marks)

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12. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ be four position vectors in 3D space such that their lengths are all equal to 1, i.e., $|\mathbf{v}_1| = |\mathbf{v}_2| = |\mathbf{v}_3| = |\mathbf{v}_4| = 1$. The terminal points of these four vectors are V_1, V_2, V_3, V_4 respectively, forming a tetrahedron.

(a) (i) Prove that $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) = \mathbf{v}_1 \times \mathbf{v}_2 + \mathbf{v}_2 \times \mathbf{v}_3 + \mathbf{v}_3 \times \mathbf{v}_1$. (2 marks)

(ii) If $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3) \neq (\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_4)$, explain why the four points V_1, V_2, V_3, V_4 are not coplanar. (1 mark)

(b) Now assume $\mathbf{v}_1 = \mathbf{k}$, $\mathbf{v}_2 = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$, $\mathbf{v}_3 = -\cos \theta \mathbf{i} + \sin \theta \mathbf{j}$, and $\mathbf{v}_4 = -\mathbf{k}$, where $0 < \theta < \frac{\pi}{2}$.

(i) Calculate $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_3)$ and $(\mathbf{v}_1 - \mathbf{v}_2) \times (\mathbf{v}_1 - \mathbf{v}_4)$, and verify that they are not equal. (2 marks)

(ii) Find the volume V of the tetrahedron $V_1V_2V_3V_4$ in terms of θ . (2 marks)

(c) As θ varies,

(i) find the maximum volume of the tetrahedron $V_1V_2V_3V_4$. (2 marks)

(ii) when the volume attains its maximum value, find the dihedral angle between the face $V_1V_2V_3$ and the base $V_2V_3V_4$. (3 marks)

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